

AML — Adverse-Risk Genetics



1. ADVERSE banner

WHAT YOU SEE

A weathered ADVERSE banner strung across the dark red cavern entrance — the blood-red cave marks this as the worst-prognosis ELN risk group.

WHAT IT ENCODES

Per ELN 2022 risk stratification, these genetic lesions define adverse-risk AML, which carries the worst overall survival (~10-20% at 5 years with chemo alone). Adverse risk is the prime indication for allogeneic HCT in first complete remission (CR1) in eligible patients, because chemotherapy alone rarely cures this group. Risk drives the entire post-remission strategy: favorable → chemo consolidation, intermediate/adverse → transplant.

BOARD TRAP

Risk group is assigned by genetics, NOT by morphology or blast count; an adverse-risk patient with 25% blasts still needs allo-HCT planning even if they achieve CR.

2. FLT3-ITD

WHAT YOU SEE

A glowing FLT3 electrical transformer wrapped in ITD coils that crackle and throw sparks — the live, sparking coils are the constitutively active kinase driving proliferation.

WHAT IT ENCODES

FLT3 internal tandem duplication (ITD) is a constitutively active receptor tyrosine kinase driving proliferation; it occurs in ~25-30% of AML and confers a high relapse rate. Add midostaurin to 7+3 induction and consolidation for newly diagnosed FLT3-mutated AML (RATIFY trial); quizartinib is also approved with chemo for FLT3-ITD. High allelic ratio (≥0.5) is more strongly adverse than low ratio.

BOARD TRAP

In ELN 2022 the FLT3-ITD allelic ratio no longer changes the risk category by itself — FLT3-ITD now places the patient in the INTERMEDIATE group (high ratio is not automatically adverse); the classic teaching that high-ratio FLT3-ITD is adverse is the older ELN 2017 rule. Also, FLT3 is NOT favorable even when NPM1 is mutated.

3. R/R relapsed-refractory

WHAT YOU SEE

Two skeleton keys tagged R/R, stamped MIDO and GILT — the R/R keys 'unlock' the relapsed/refractory setting where gilteritinib is the answer.

WHAT IT ENCODES

In relapsed/refractory FLT3-mutated AML (ITD or TKD), gilteritinib monotherapy is the standard (ADMIRAL trial: superior survival vs salvage chemo). Distinguish from the upfront setting where midostaurin/quizartinib are given WITH chemo.

BOARD TRAP

Gilteritinib covers both FLT3-ITD and FLT3-TKD point mutations; midostaurin also covers TKD, but quizartinib is ITD-selective and does not target TKD.

4. TP53 mutation

WHAT YOU SEE

An exhausted TP53 figure slumped lifeless in an armchair — the collapsed guardian is the lost p53 tumor-suppressor, the single worst lesion in AML.

WHAT IT ENCODES

TP53 mutation is the single worst prognostic lesion in AML — median survival ~5-9 months, high chemoresistance, and frequent association with complex/monosomal karyotype and therapy-related/secondary AML. Even allo-HCT rarely overcomes it. Often treated with hypomethylating agent (azacitidine) ± venetoclax rather than intensive 7+3 because intensive chemo offers little benefit.

BOARD TRAP

TP53-mutated AML responds poorly to standard 7+3 and relapses early even after transplant; do not assume achieving CR equals cure here.

5. High VAF

WHAT YOU SEE

A skull-marked balloon reading VAF floating above the slumped TP53 figure — the high-riding balloon is a high variant allele frequency / multi-hit TP53.

WHAT IT ENCODES

For TP53, a variant allele frequency (VAF) ≥10% and/or biallelic (multi-hit) involvement defines the adverse, chemoresistant phenotype. Multi-hit TP53 is now recognized as its own AML/MDS entity in WHO/ICC classifications.

BOARD TRAP

A low-VAF single TP53 mutation may represent clonal hematopoiesis rather than driving disease; the prognostic weight is greatest with multi-hit/high-VAF TP53.

6. -5 / -7 del

WHAT YOU SEE

Torn chromosome cards stamped -5 and -7 scattered across the cave floor — the missing/ripped chromosomes are monosomy 5/7 and complex karyotype.

WHAT IT ENCODES

Loss of chromosome 7 (-7), del(5q)-5, abn(17p), and complex karyotype (≥3 unrelated abnormalities) or monosomal karyotype are all ELN adverse. These are hallmarks of myelodysplasia-related and therapy-related AML.

BOARD TRAP

-7 (monosomy 7) is adverse, but isolated trisomy 8 alone is intermediate, not adverse; do not lump all numerical abnormalities together.

7. KMT2A (MLL) 9;11

WHAT YOU SEE

A heavy padlocked door labeled KMT2A with a 9;11 tag hanging on it — the locked gate is the 11q23 rearrangement that blocks normal maturation.

WHAT IT ENCODES

KMT2A (MLL, 11q23) rearrangements are adverse, classically t(9;11) MLLT3::KMT2A and other partner fusions; strongly linked to therapy-related AML after topoisomerase-II inhibitors (etoposide, anthracyclines).

BOARD TRAP

t(9;11) is the LEAST aggressive KMT2A partner and is sometimes labeled intermediate, but ELN groups KMT2A rearrangements as adverse overall — and do not confuse the leukemogenic t(9;11) with benign findings.

8. ASXL1

WHAT YOU SEE

A mugshot card reading ASXL1 in the criminal lineup along the bottom — the lineup of 'repeat offenders' is the secondary-type / MDS-related mutation panel.

WHAT IT ENCODES

ASXL1 is a secondary-type / myelodysplasia-related mutation; its presence defines adverse risk and suggests AML arising from antecedent MDS/MPN. Part of the secondary-type mutation signature (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2).

BOARD TRAP

Any single secondary-type mutation (e.g. ASXL1) flags the AML as myelodysplasia-related and adverse, regardless of an otherwise normal karyotype.

9. BCOR

WHAT YOU SEE

A mugshot card reading BCOR in the bottom lineup — another 'repeat offender' secondary-type mutation flagging MDS-derived, adverse biology.

WHAT IT ENCODES

BCOR is a secondary-type/myelodysplasia-related mutation conferring adverse risk, again pointing to a secondary or MDS-derived AML biology.

10. EZH2

WHAT YOU SEE

A mugshot card reading EZH2 in the bottom lineup — a secondary-type adverse mutation in the same MDS-related rogues' gallery.

WHAT IT ENCODES

EZH2 is a secondary-type adverse mutation (polycomb repressive complex 2 member), part of the myelodysplasia-related signature.

11. RUNX1

WHAT YOU SEE

A mugshot card reading RUNX1 in the bottom lineup — the criminal-lineup RUNX1 is the adverse point MUTATION, opposite to the favorable t(8;21) fusion.

WHAT IT ENCODES

A somatic RUNX1 mutation (point mutation/indel) is adverse and is a secondary-type/myelodysplasia-related lesion. This is mechanistically and prognostically opposite to the favorable t(8;21) RUNX1::RUNX1T1 FUSION.

BOARD TRAP

Mutated RUNX1 = adverse; the t(8;21) RUNX1::RUNX1T1 translocation = favorable core-binding-factor AML. Same gene name, opposite prognosis — a classic board trap.

12. Splice (SF3B1/SRSF2/U2AF1/ZRSR2)

WHAT YOU SEE

A row of figures under a SPLICE sign (SF3B1, SRSF2, U2AF1, ZRSR2) — the spliceosome crew are the MDS-related secondary-type mutations that pull AML into adverse risk.

WHAT IT ENCODES

Spliceosome mutations (SRSF2, U2AF1, ZRSR2; SF3B1 in this context) are myelodysplasia-related secondary-type mutations conferring adverse risk and indicating MDS-derived biology.

BOARD TRAP

SF3B1 is classically the GOOD-prognosis mutation in MDS with ring sideroblasts, but in AML it falls into the adverse secondary-type group — context flips its meaning.

13. IDH

WHAT YOU SEE

Glowing potion bottles labeled IDH lined up on a shelf with a goblin — the brewable/targetable potions are IDH1/2, druggable but only intermediate risk.

WHAT IT ENCODES

IDH1 and IDH2 mutations produce the oncometabolite 2-hydroxyglutarate, causing a hypermethylation/differentiation block. They are TARGETABLE: ivosidenib for IDH1, enasidenib (or ivosidenib) for IDH2; olutasidenib for IDH1. IDH mutation alone is intermediate risk, not adverse.

BOARD TRAP

IDH inhibitors can cause differentiation syndrome (like ATRA in APL), treated with steroids. IDH mutation is NOT adverse by itself — it is placed here as a targetable lesion, intermediate risk.

14. TKD ? panel

WHAT YOU SEE

A meter panel reading TKD with a big question mark beside the FLT3 transformer — the uncertain '?' gauge is FLT3-TKD, whose adverse weight is unclear (not equal to ITD).

WHAT IT ENCODES

FLT3-TKD (tyrosine kinase domain, D835) point mutations are distinct from FLT3-ITD; their prognostic impact is less clearly adverse and they do not by themselves confer the adverse risk that ITD does. They are still targetable by midostaurin and gilteritinib (but NOT quizartinib).

BOARD TRAP

FLT3-TKD is NOT prognostically equivalent to FLT3-ITD; quizartinib does not cover TKD mutations whereas gilteritinib and midostaurin do.

15. Adverse outcome clock

WHAT YOU SEE

A wall clock ticking beside the slumped TP53 figure — the running clock is the short time-to-relapse that forces urgent donor search and CR1 transplant.

WHAT IT ENCODES

Adverse-risk genetics drive short time-to-relapse and poor overall survival, which is why these patients are referred urgently for donor search and allo-HCT in CR1 rather than relying on chemotherapy consolidation alone.

BOARD TRAP

Delaying donor identification in adverse-risk AML loses the narrow CR1 window for transplant — the chance of cure falls sharply once the disease relapses.